ANIL PANCHITHA

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PROFESSIONAL SUMMARY

Software Engineer with **1 year of experience at Cisco Systems**, specializing in **Python, Generative AI, LLMs, RAG, Machine Learning, and AI-driven automation**. Hands-on experience building **RAG applications, LLM solutions, API integrations, and automation workflows** for enterprise environments. Worked on solutions supporting **Hilton and Wells Fargo**, with experience in **embeddings, vector databases, retrieval, reranking, RAG evaluation, REST APIs, and ML**.

TECHNICAL SKILLS

Programming	Python, SQL
AI/ML & GenAI	Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Prompt Engineering
LLM Tools	LangChain, OpenAI API, ChromaDB
Frameworks	TensorFlow, Scikit-learn
APIs & Tools	REST APIs, FastAPI, Git, Postman, Pandas, NumPy

EXPERIENCE

Software Engineer | Cisco Systems | Bangalore, India **Aug 2025 – Aug 2026**

- Contributed to the development of a **GenAI-powered CMS AI Assistant** using **LLMs, RAG, prompt engineering, and enterprise knowledge retrieval** to assist engineers with troubleshooting and operational information.
- Designed and integrated **RAG pipelines** for enterprise knowledge, including document processing, chunking, embeddings, vector search, context retrieval, and grounded LLM response generation.
- Developed **Python and REST API-based automation workflows** integrating Cisco Meraki APIs and ServiceNow to automate recurring network operations and reduce manual effort.
- Built and supported automation solutions for enterprise customers including **Hilton and Wells Fargo**, implementing pre-validation, configuration, post-validation, reporting, and notification workflows.
- Applied **Python, SQL, APIs, JSON, and data processing** to troubleshoot production issues, analyze operational data, and develop reusable engineering solutions.
- Worked on **AI-assisted automation and knowledge-retrieval use cases**, improving access to technical information and reducing repetitive manual effort for engineering teams.
- Collaborated with cross-functional engineering teams to perform **root-cause analysis, debugging, and production support**, delivering reliable automation and AI-assisted solutions.

PROJECTS

FarmerVerify AI – AI-Powered Document Verification System **2026**

Python, LLMs, RAG, FastAPI, GPT-4.1 Vision, NLP, OCR, Embeddings, Vector Search, REST APIs

- Built a **Generative AI-powered document verification platform** that reduced onboarding time from **14 days to less than 2 hours** through automated document processing and validation.
- Developed a **RAG pipeline** using embeddings, vector search, and LLMs for information retrieval, question answering, and document validation.
- Integrated **GPT-4.1 Vision and multimodal AI** to process regional-language documents and unstructured image-based records.
- Designed an **8-stage automated workflow** covering OCR, preprocessing, fuzzy matching, deduplication, validation, fraud detection, and decision support.
- Built **FastAPI-based backend services and REST APIs** to orchestrate document processing, AI inference, and workflow automation.
- Applied **prompt engineering** to improve information extraction, document understanding, and consistency of LLM-generated responses.

CMS AI Assistant – Enterprise GenAI Knowledge Assistant

Python, LangChain, LLMs, RAG, ChromaDB, Embeddings, NLP, Prompt Engineering, Vector Search

- Developed an **LLM-powered enterprise knowledge assistant** for Cisco CMS engineers to retrieve technical information and support troubleshooting workflows.
- Built an end-to-end **RAG pipeline** covering document ingestion, preprocessing, chunking, embeddings, vector indexing, semantic retrieval, and LLM-based response generation.

- Implemented **ChromaDB-based vector search** with metadata filtering and similarity search to improve retrieval accuracy.
- Added a **reranking layer** to improve document ordering and reduce irrelevant context passed to the LLM.
- Evaluated and optimized the RAG pipeline using **Precision@K, Recall@K, MRR, faithfulness, chunk size, top-K retrieval, and similarity thresholds**.

EDUCATION

Institute of Aeronautical Engineering Hyderabad, India

B.Tech in Data Science *CGPA: 7.34* *2021 – 2024*

PUBLICATIONS

Diabetic Retinopathy Detection Using Deep Learning **Springer – ICATECS 2024/2025**

- Published research on diabetic retinopathy detection using **EfficientNet-B1, Transfer Learning, and the IDRiD dataset**, with comparative evaluation against **ResNet50 and InceptionResNetV2**.
- Applied **Grad-CAM** for model explainability and published in **Springer LNICST, Volume 620**. **DOI:** 10.1007/978-3-031-94283-9_21 Springer Link

CERTIFICATIONS

- **NPTEL Data Science Domain Scholar**
- Machine Learning Specialization – Coursera
- Google Data Analytics Professional Certificate
- IBM Data Science Professional Certificate

ACHIEVEMENTS

- **Winner – Cisco Tech Hackathon 2026 (You Amaze Award)**
- Featured on Cisco's LinkedIn and Cisco's X (Twitter).
- Recognized by Executive CX Leadership for innovation and operational excellence.